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Scrimbal Assignment Graded Quiz Vector Databases for Recommendation Systems and RAG
Graded Assignment, 15 min

Due: Aug 24, 11:59 PM IST

Your grade: 100%

You received **100%**. Our highest **100%**. To pass you need at least 70% Retake your highest score.

Retake assignment

Attempt number this assignment: **1/3**

Applied Generative & Development and Strategy

YOUR GRADE
100%

1. **How does a vector database facilitate the implementation of an recommendation system?** 1 / 1 point

   - By optimizing SQL queries for faster retrieval times
   - By storing and indexing embedding of important item features
   - By generating new data points based on existing user data
   - By directly storing user preferences as text blocks

   Nice work

   A vector database stores embeddings which encapsulate representation features, enabling similarity searches essential for recommendation systems.

2. **What is a key advantage of using a vector database for Retrieval Augmented Generation (RAG)?** 1 / 1 point

   - Frequently wait time delays resulting from large datasets.
   - Inability to efficiently perform similarity searches using vector embeddings.
   - Enabled the LLMs to deliver augmented responses from the original base prompt and the retrieved text.
   - Expensive advanced visualization tools for data analysis.

   Nice work

   Vector databases enhance RAG by enabling efficient similarity searches, which are crucial for retrieving relevant information using vector embeddings.

3. **In the context of vector databases, how can similarity searches be applied to improve user experience in recommendation systems?** 1 / 1 point

   - By identifying items that are closely related in feature spaces to the user's preferences
   - By using SQL queries to filter out irrelevant data
   - By generating random recommendations based on user history
   - By storing user data in a structured tabular format

   Nice work

   Similarity searches in vector databases identify items that are closely related in feature spaces, which helps in providing personalized recommendations.

4. **How do embedding models and vector databases work together in a recommendation system?** 1 / 1 point

   - Embedding models replace vector databases to simplify data storage
   - Embedding models visualize data relevant to vector databases
   - Vector databases analyze data to create embeddings for recommendations
   - Vector databases store embeddings that are used for similarity searches.

   Nice work

   Embedding models generate vectors that are stored in vector databases, facilitating similarity searches for recommendations.

5. **Which of the following tasks can be done after modify / reshape a Chroma collection?** 1 / 1 point

   - Change the distance metric
   - Change the collection metadata
   - Change the collection name
   - Add a key-value pair to the collection metadata

   Nice work

   In Chroma DB, the distance metric must be specified when the collection is created. To change it later, you need to clone the collection using the modify method and then work.

6. **What is a common pitfall in a data pipeline?** 1 / 1 point

   - Choosing a chunk size that is too large (enough to keep the meaning) but
   - Re-embedding data after changes
   - Using a different embedding model for query and document embedding
   - Using vector databases for storage

   Nice work

   Using a different embedding model for query and document embedding will typically lead to poor retrieval performance and is a critical blunder.

7. **Which of the following RAG tasks is typically handled outside the vector database?** 1 / 1 point

   - Chunking
   - Embedding prompts
   - Storing embeddings
   - Retrieving relevant matches

   Nice work

   Chunking is typically performed outside the vector database, with the resulting documents being incorporated into the vector database for embedding and storage.